

What has happened to human fertility?



Semen quality appears to have declined in recent decades in some populations, e.g. north-western Europe. At the same time, couple fertility may have increased. Hypotheses are suggested for this apparent inconsistency. Alongside the deterioration of spermatogenesis there is clear evidence of an increase in other related problems, notably testicular cancer. The sharply rising trend in this condition started a century ago--decades earlier than sometimes thought. This and other evidence clearly indicates an environmental origin, but there is also a definite genetic component. The relationship of genetics and environment is discussed in the context of the puzzle that infertility is inherited, which appears to be impossible from an evolutionary standpoint. Poor semen quality is related not only to testicular cancer but also to zygote development, in which cancer-like disruption of the genetic apparatus is observed, with serious implications for offspring health. This needs to be seen in the context that human reproduction is prone to a higher degree of impairment than that of other mammalian species, in relation to spermatogenesis, couple fertility, early pregnancy loss and embryonic aneuploidy; female- and male-mediated pathways are both implicated. It is unclear whether such human specificity originated on an evolutionary/genetic or a historico-social timescale, which is important in relation to pathogenesis. The evidence clearly indicates that the currently most popular explanation for male reproductive system impairment, the endocrine disruption hypothesis, cannot explain the main features of the descriptive epidemiology. An alternative pathogenesis is outlined, and some possible exposures considered that could be responsible.